

Name: Tejasvi M

Reg no: 192511047

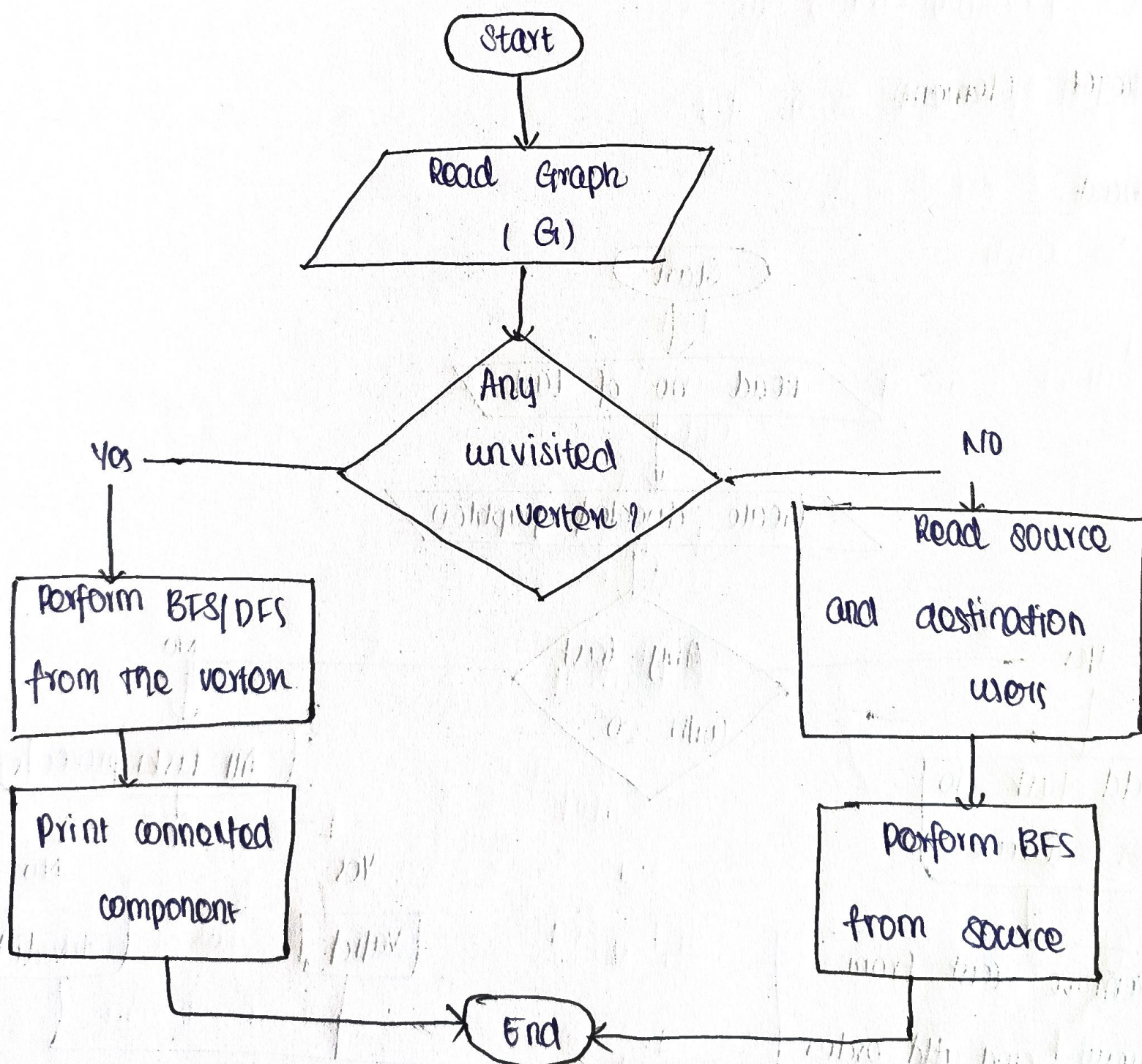
Course: Data Structures

Date: 30/08/24

Q5 - AT3 - Flowchart and code

1) Array traversal

Flowchart:



code:

```
#include <stdio.h>
```

```
int main() {
```

```
int a[100], n, i;
```

```
scanf("%d", &n);
```

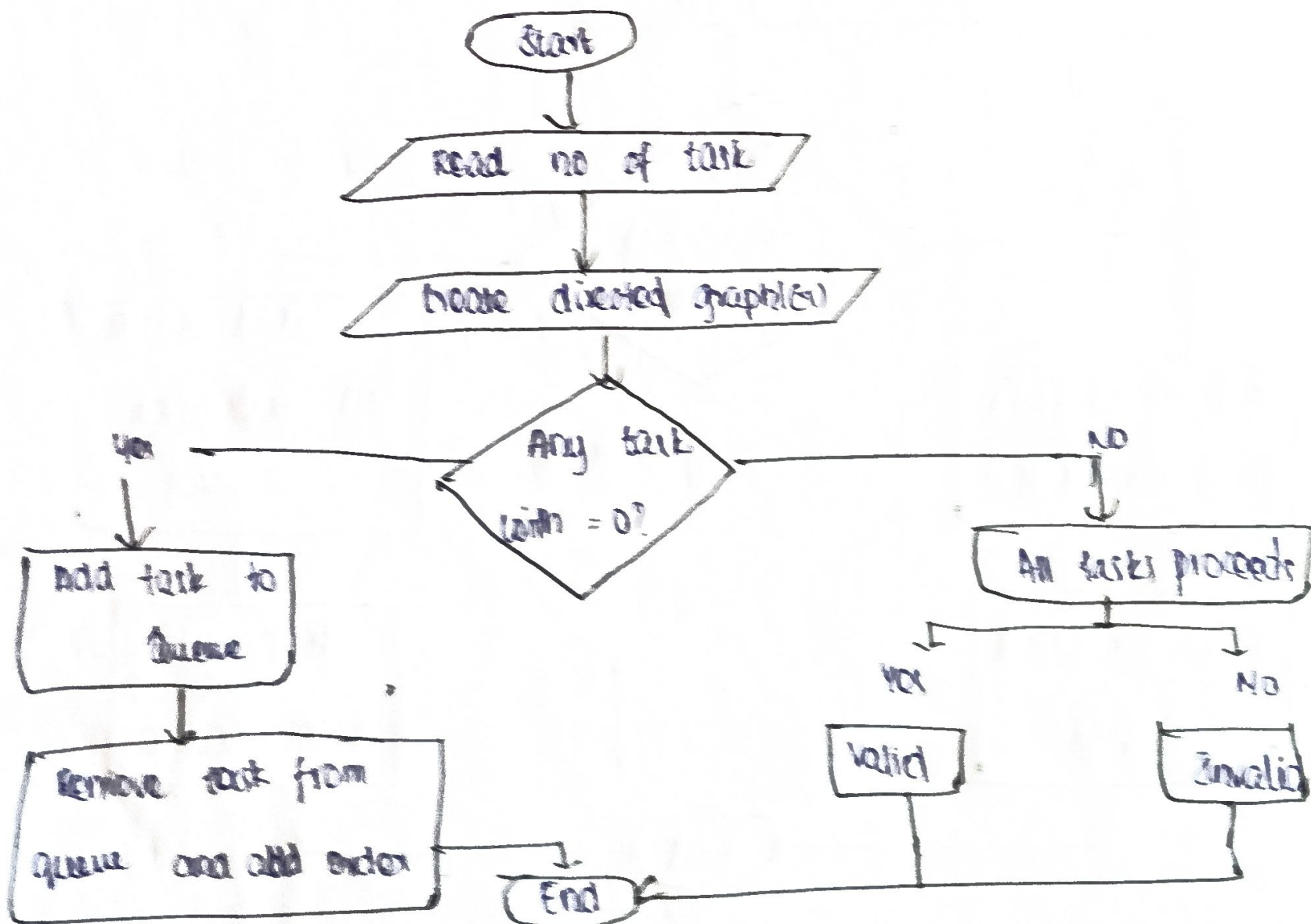
```
for (i=0; i<n; i++) scanf("%d", &a[i]);
```

```
for (i=0; i<n; i++) printf("%d", a[i]);
```

```
}
```

2) Target Element

Flowchart



code.

```
#include <stdio.h>
```

```
int main()
```

```
{  
    int a[100], n, i, max;
```

```
    scanf("%d", &n);
```

```
    for (i=0; i<n; i++) scanf("%d", &a[i]);
```

```
    max = a[0];
```

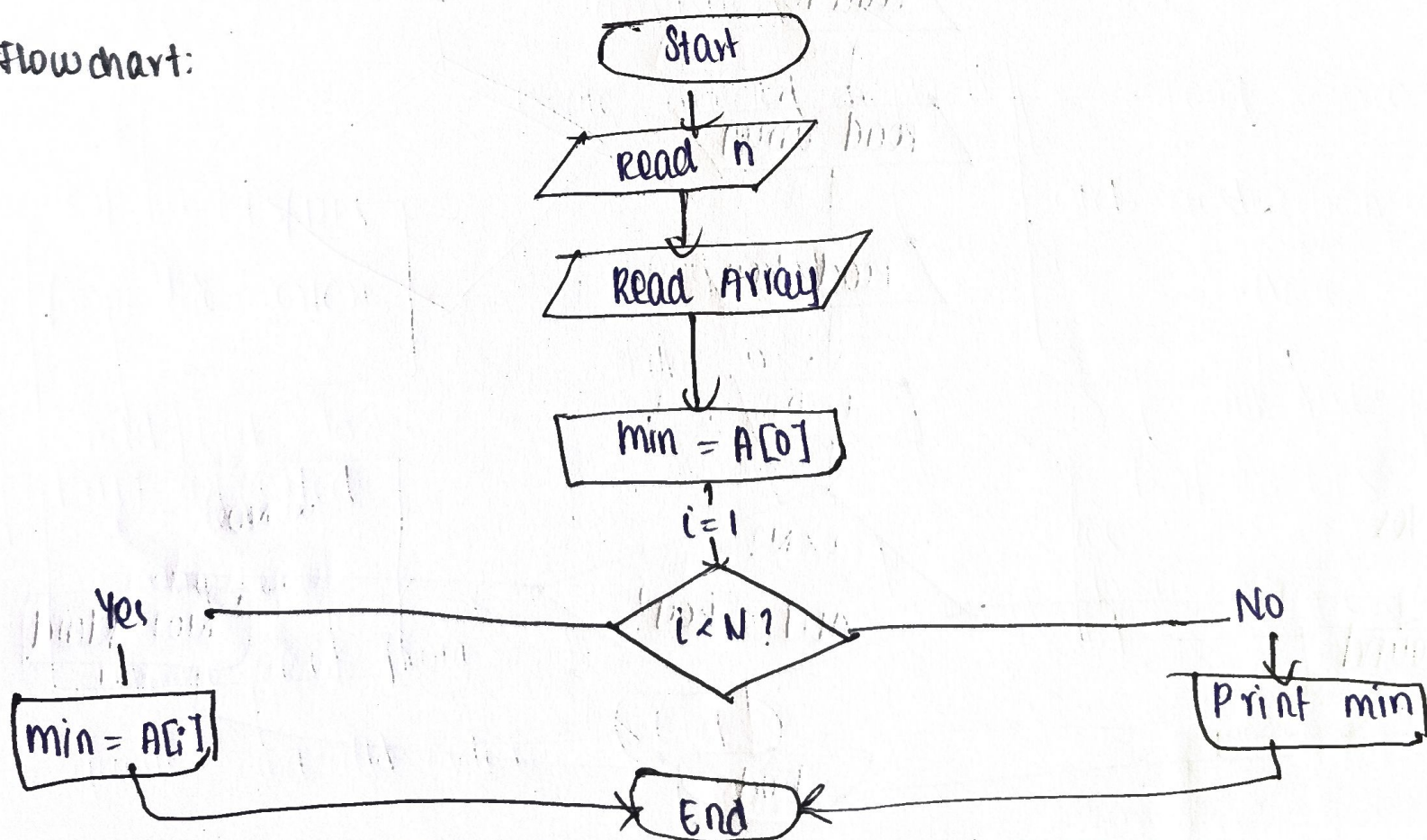
```
    for (i=1; i<n; i++)
```

```
        if (a[i] > max) max = a[i];
```

```
    printf("%d", max);  
}
```

3) Smallest in Array

Flowchart:



code:

```
#include <stdio.h>
```

```
int main () {
```

```
int a[100], n, i, min;
```

```
scanf ("%d", &n);
```

```
for (i=0 ; i<n; i++) scanf ("%d", &a[i]);
```

```
min = a[0];
```

```
for (i=1 ; i<n; i++)
```

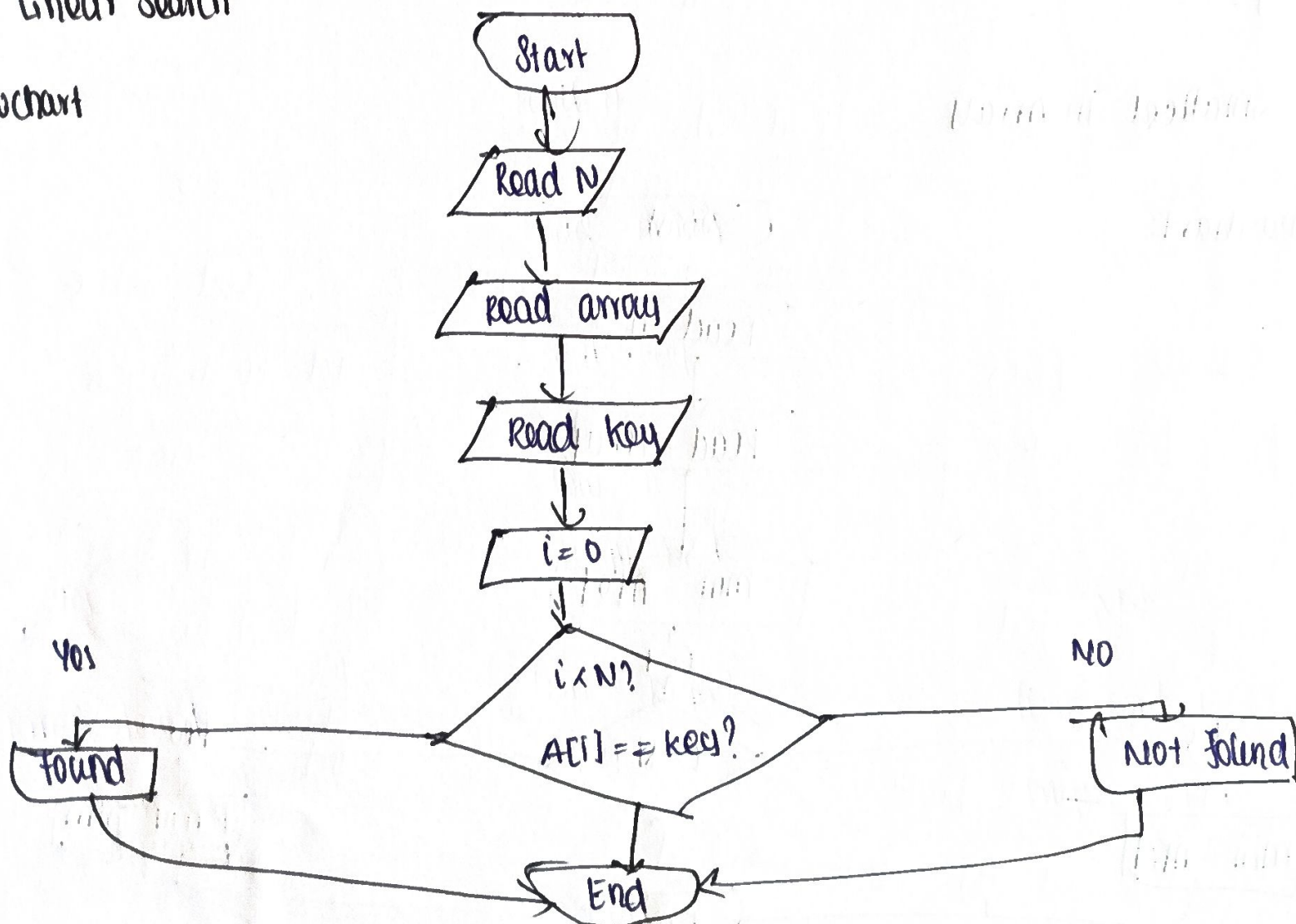
```
if (a[i] < min) min = a[i];
```

```
printf ("%d", min);
```

4

4.) Linear Search

Flowchart



code:

```
#include <stdio.h>
```

```
int main () {
```

```
int a[100], n, i, key, found = 0;
```

```
scanf ("%d", &n);
```

```
for (i = 0; i < n; i++) scanf ("%d", &a[i]);
```

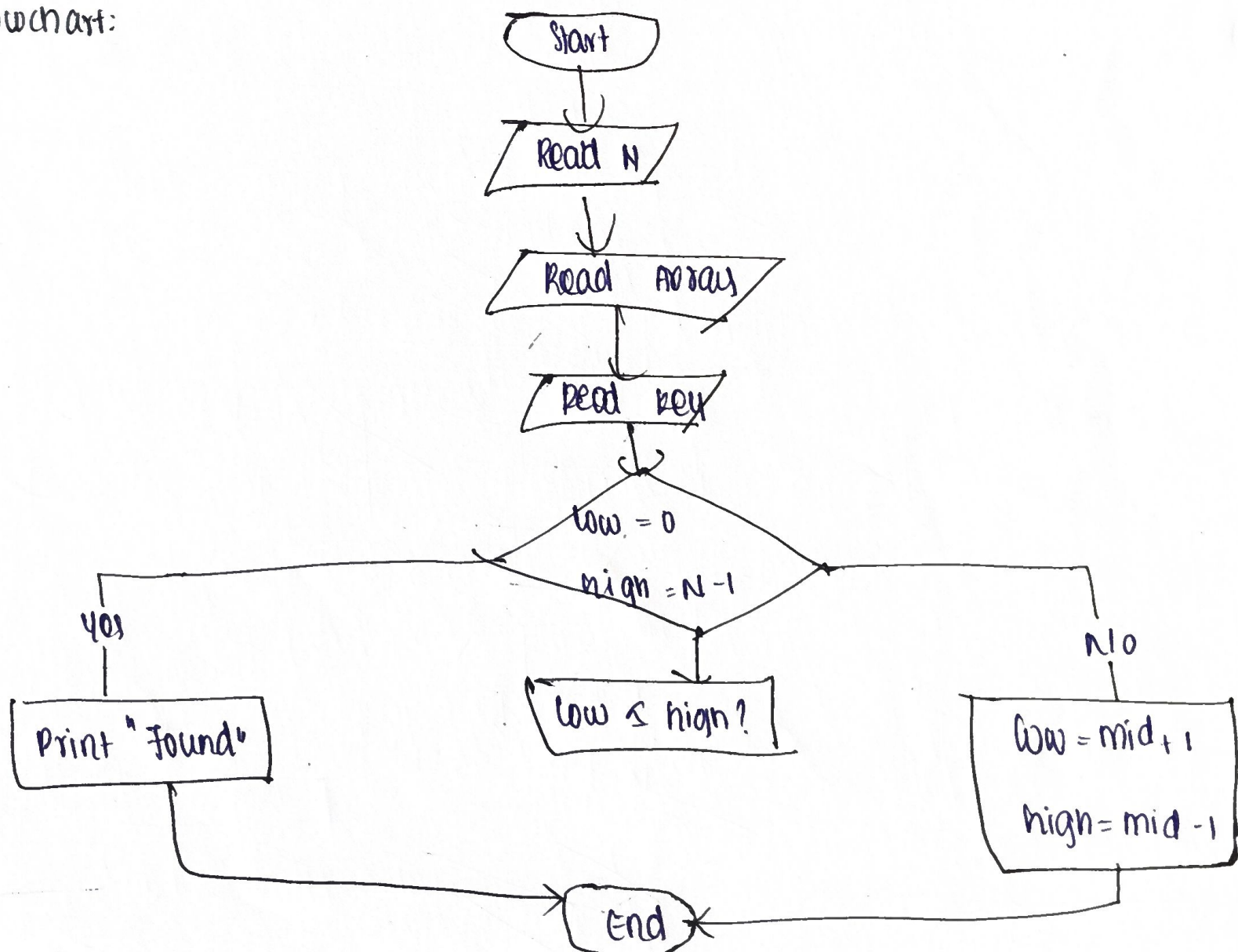
```
scanf ("%d", &key);
```

```
else printf ("Not found");
```

4

5. Binary Search

Flowchart:



code.

```
#include <stdio.h>
```

```
int main()
```

```
{ int a[100], n, i, key, i = 0, n, m;
```

```
scanf ("%d", &n);
```

```
for (i = 0; i < n; i++)
```

```
scanf ("%d", &a[i]);
```

```
if (a[m] == key)
```

```
printf ("Found");
```

```
}
```

```
printf ("Not found");
```

```
}
```

